Specifications

General Considerations

The drawings are diagrammatic and do no show exact locations. The HVAC subcontractor shall determine and coordinate the installed locations and elevations of all equipment with the Owner and other subcontractors.

Refrigerant tubing, condensate piping, ductwork, electric conduit, etc. shall be installed neatly and generally parallel to the building structure wherever possible. Refrigerant tubing shall be installed as one continuous piece from the outdoor to the indoor unit - there shall be no concealed fittings.

See Drawing H-1 for a schedule of proposed equipment. Other drawings included in the scope of work include H-2, H-3, H-4 and H-5.

All airflow systems shall be commissioned, including the ERV and the AHUs. Air flow rate and static pressure measurements shall be recorded on a schematic for the system. Results shall be reported to the Owner.

All equipment shall be installed as per manufacturer recommendations and with strict adherence to service clearances.

All equipment shall be installed in accordance with the MUBEC, see https://www.maine.gov/dps/fmo/building-codes.

Applications for Efficiency Maine incentives shall be completed on behalf of the Owner by the HVAC subcontractor.

Heating and Cooling Systems

Heat Pump Indoor Unit

General

The heat pump air conditioning system shall be a Mitsubishi Electric M-Series variable capacity, multi-position, air-handling unit and matching inverter, cold-climate-rated, outdoor unit, or approved equal. The system shall consist of a combination of an indoor unit with a remote controller connected to a compact horizontal discharge outdoor unit which shall be of an inverter driven heat pump design.

The system installation shall comply with all manufacturer requirements.

Indoor Unit: Multi-position Air Handling Unit

Mitsubishi, SVZ Series, or equal. The multi-position indoor unit shall be factory assembled, wired and run tested. Contained within the unit shall be all factory wiring, piping, electronic modulating linear expansion device, control circuit board and fan motor. The unit shall have a self-diagnostic function, 3-minute time delay mechanism, and an auto restart function. Indoor unit and refrigerant pipes shall be charged with dehydrated air before shipment from the factory. The unit shall be suitable for use in air handling spaces in accordance with Section 18.2 of UL 1995 4th Edition, be tested in accordance with ANSI/ASHRAE 193 and have less than 2% air leakage at maximum airflow setting.

Unit Cabinet

The cabinet shall include a fixed bottom return, a fixed vertical discharge supply and be pre-painted, pre-insulated, 22 gauge galvanized steel or utilize black ZAM steel.

Fan

The indoor unit fan shall be an assembly with a single, statically and dynamically balanced direct drive fan with a high efficiency DC motor with permanently lubricated bearings.

The fan shall have 3-speeds with the capability to operate between 0.3, 0.5, or 0.8 in. WG, selectable of external static resistance.

Filter

The unit shall have a 1” filter rack with a reusable filter.

Accessory Filter Section

An accessory filter section shall be included. The section shall be a minimum of 6” of depth to accommodate a 5” deep filter. The filter shall be non-woven, 5” thick, pleated with efficiency of Merv-11. It shall present a maximum initial resistance (300 fpm) of 0.15 inches WG. Air Bear™, or equal.

Cooling and Heating Coil

The indoor unit coil shall be of nonferrous construction with smooth plate fins on copper tubing. The tubing shall have inner grooves for high efficiency heat exchange. All tube joints shall be brazed with phos-copper or silver alloy.

The coils shall be pressure tested at the factory.

Electrical

The unit electrical power service shall be 240 Volts, 1-phase, 60 hertz.

The system shall be equipped with a system directing that the indoor unit be powered directly from the outdoor unit using a 3-wire, 14-gauge AWG connections plus ground.

Optional Electric Heat Kit

The indoor unit shall have a manufacturer supplied electric heat kit accessory. The electric heat kit shall offer either one or two stages of back up heat for maximum efficiency. The heater shall be designed to work with the indoor unit without any modifications to the unit or to the control sequence.

The heater shall be powered from a dedicated electrical feed, not from the indoor unit.

Controls

Control board shall include contacts for control of no less than two stages of external heat. The first stage of external heat may be energized when the space temperature is 2.7°F from set point for between 10-25 minutes (user adjustable). The second stage of external heat may be energized when the first stage has been active for no less than 5 minutes and the space temperature has not risen by more than 0.9°F.

Refrigerant Line-sets

Line sets shall be Mueller Streamline™, Ebrille Coveral™, or equal. Line sets tubing shall be insulated with elastomeric insulation or foamed polyethylene with an anti-UV outer aluminum film cover, minimum 0.5 in. thickness . Line Sets shall be third party verified through Underwriters Laboratories (UL) for a minimum operating pressure of 700 psig at 250°F. This rating includes fully annealed and brazed copper tubing.

PDM Gelcopper™ line sets are not approved.

Condensate Piping

Condensate piping shall be rigid PVC, ¾” minimum size, schedule 40.

Electric Wire

Indoor units wire shall be Honeywell - 14/4 Stranded THHN 600V Tray Cable Black Minisplit Wire, or equal.

The indoor unit shall have the option to be powered independently from the outdoor unit.

Heating and Cooling Ductwork

Steel ductwork

Trunk ductwork shall be rigid steel, 26 gage, or approved equal.

Longitudinal and transverse joints, seams and connections shall be sealed air-tight

Supply and return ductwork shall be insulated with a minimum of R-6 wrap.

Ducts that penetrate the residence air-barrier shall be sealed at the air barrier.

Flexible ductwork

Flexible ductwork shall be ATCO, or equal.

Flexible ductwork shall be limited to runouts of less than or equal to 10 feet. Runout duct may be flexible when 10 feet or less if installed to minimize static pressure drop.

Air Terminal Units

Supply and return air terminal units shall be Hart and Cooley, or equal.

Coordinate with the Architect and General Contractor on for color and appearance. See H-1 for the schedule.

Controls

Heat pump indoor units

Heat pump indoor units shall be controlled by local wall thermostats and/or smartphone apps.

Wall thermostats shall be 7-day programmable, include relative humidity and temperature sensors

Wall thermostat shall be capable of wireless communication with indoor units, Mitsubishi, MK2, or equal.

WIFI adapters shall be install for all indoor units. Control shall be available from Smartphone apps such Mitsubishi kumo Cloud™, or equal.

Remote access and control of the heat pump systems shall be available off-site.

Cold Climate Heat Pump Outdoor Unit

General

Mitsubishi, SUZ Series, or equal. The outdoor units shall have a thermally fused powder coated finish. The outdoor unit shall be completely factory assembled, piped and wired. Each unit shall be run tested at the factory.

Outdoor unit shall have a sound rating no higher than 55 dB(A). If an alternate manufacturer is selected, any additional material, cost, and labor to meet published sound levels shall be incurred by the contractor.

Refrigerant lines from the outdoor unit to the indoor units shall be insulated in accordance with the installation manual.

The outdoor unit shall meet performance requirements per schedule and be within piping limitations & acceptable ambient temperature ranges as described in respective manufacturers' published product catalogs. Non-published product capabilities or performance data are not acceptable.

The outdoor unit shall be capable of guaranteed operation in heating mode down to -13 °F ambient temperatures and cooling mode up to 115 ° F without additional restrictions on line length & vertical separation beyond those published in respective product catalogs.

Unit Cabinet

The casing shall be fabricated of galvanized steel, bonderized, finished with an electrostatically applied, thermally fused acrylic or polyester powder coating for corrosion protection. Assembly hardware shall be cadmium plated for weather resistance.

Two (2) mild steel mounting feet, traverse mounted across the cabinet base pan, welded mount, providing four (4) slotted mounting holes shall be furnished. Assembly shall withstand lateral wind gust up to 155 MPH to meet applicable weather codes. The casing(s) shall be fabricated of galvanized steel, bonderized and finished.

Fan

The unit shall be furnished with a direct drive propeller type fan. The outdoor unit fan motor shall be a electrically commutated, direct current (DC), constant torque motor and have permanently lubricated bearings. The fan motor shall be mounted for quiet operation. The fan shall be provided with a raised guard to prevent contact with moving parts. The outdoor unit shall have horizontal discharge airflow.

Refrigerant and Refrigerant Piping

R410A refrigerant shall be required for systems. Polyolester (POE) oil shall be required.

Refrigerant piping shall be phosphorus deoxidized copper (copper and copper alloy seamless pipes) of sufficient radial thickness as defined by the equipment manufacturer and installed in accordance with manufacturer recommendations.

All refrigerant piping must be insulated with ½” closed cell, CFC-free foam insulation with flame-Spread Index of less than 25 and a smoke-development Index of less than 50 as tested by ASTM E 84 and CAN / ULC S-102. R value of insulation must be at least 3.

Coil

The outdoor unit coil shall be of nonferrous construction with lanced or corrugated plate fins on copper tubing.

The coil fins shall have a factory applied corrosion resistant Blue Fin finish. Uncoated aluminum coils/fins are not allowed.

The coil shall be protected with an integral metal guard.

Refrigerant flow from the outdoor unit shall be regulated by means of an electronically controlled, precision, linear expansion valve.

Outdoor unit shall be pre-charged with sufficient R-410a refrigerant for up to twenty-five (25) feet of refrigerant piping for capacities up to 24,000 BTU/h.

All refrigerant lines between outdoor and indoor units shall be of annealed, refrigeration grade copper tubing, ARC Type, meeting ASTM B280 requirements, individually insulated in twin-tube, flexible, closed-cell, CFC-free (ozone depletion potential of zero), elastomeric material for the insulation of refrigerant pipes and tubes with thermal conductivity equal to or better than 0.27 BTU-inch/hour per Sq Ft / °F, a water vapor transmission equal to or better than 0.08 Perm-inch and superior fire ratings such that insulation will not contribute significantly to fire and up to 1” thick insulation shall have a Flame-Spread Index of less than 25 and a Smoke-development Index of less than 50 as tested by ASTM E 84 and CAN / ULC S-102.

All refrigerant connections between outdoor and indoor units shall be flare type.

Compressor

The compressor shall be a high performance, hermetic, inverter driven, variable speed, dual rotary type manufactured by Mitsubishi Electric Corporation, or approved equal.

The compressor motor shall be direct current (DC) type equipped with a factory supplied and installed inverter drive package.

The compressor will be equipped with internal thermal overload protection.

The outdoor unit shall have the ability to operate over the full capacity range with a maximum height difference of 40 feet (12 meters) for 9,000, 12,000, and 15,000 BTU/h models, 50 feet (15 meters) for 18,000 BTU/h models, and 100 feet (30 meters) for 24,000 BTU/h, 30,000 BTU/h and 36,000 BTU/h models; and have refrigerant tubing length of up to 65 feet (20 meters) for 9,000, 12,000, and 15,000 BTU/h models, and 100 feet (30 meters) for 18,000 BTU/h, 24,000 BTU/h, 30,000 BTU/h and 36,000 BTU/h models between the indoor and outdoor units.

Filters, sight glasses, and traps shall not be used, and no additional refrigerant oil shall be required.

The compressor shall be mounted so as to avoid the transmission of vibration.

Base-pan Heater - Optional

If requested, each outdoor unit module shall be equipped with a base-pan heater to protect the coil against ice build-up during prolonged winter operation. Base-pan heater shall activate only if compressor is operating in heating mode at an outdoor ambient temperature of less than or equal to 36 ° F.

Electrical

The outdoor unit electrical power supply shall be 240 volts, 1-phase, 60 hertz.

The unit shall be capable of satisfactory operation within voltage limits of 187 volts to 253 volts.

The outdoor unit shall be controlled by microprocessors located in the indoor unit and outdoor unit. A 12 to 24 volt DC data stream shall communicate between the units providing all necessary information for full function control.

The outdoor unit shall be equipped with Pulse Amplitude Modulation (PAM) compressor inverter drive control.

Stand

Four-legged outdoor unit mounting stand shall be furnished and installed. The stand shall be made from 7 gauge plate steel with thermally fused polyester powder coat finish that meets ASTM D3451-06 standards. Stands shall be provided with galvanized mounting hardware and meets all ASCE 7 overturning safety requirement.

Stands shall me manufactured by Quick-Sling™, or equal, and shall elevate the outdoor unit to a minimum of 18” above grade.

Base

The stand shall be placed on a level surface capable of draining water. A surface of enclosed crushed stone will flush pavers, or equal.

Rain Shield

The outdoor unit shall be provided with a manufacturer supplied 20 gauge hot dipped galvanized snow /hail guard. The snow/hail guard protects the outdoor coil surfaces from hail damage and snow build-up in severe climates.

Shield shall be manufactured by Twin City Sheet Metal, or equal.

Electrical

Power wiring capacity shall be as required by the manufacturer.

Disconnect shall be weather-proof, 3R enclosure, knife-type, Square-D, or equal

Ventilation and Exhaust System

Energy Recovery Ventilation Unit (ERV)

Unit

The ERV manufacturer and model shall be Renewaire Premium series, or approved equal.

The Air-to-Air Energy Recovery Ventilaton (ERV) shall be fully assembled at the factory and consist of a fixed-plate cross-flow heat exchanger with no moving parts, an insulated single wall G90 galvanized painted 22-gauge steel cabinet, filter assemblies for both intake and exhaust air, enthalpy core, supply air blower assembly, exhaust air blower assembly and electrical control box with all specified components and internal accessories factory installed and tested and prepared for single-point high voltage connection. Entire unit excepting field-installed components shall be assembled and test operated at the factory.

The ERV shall use an integral mounting flange and hanging bar system to mount the unit per manufacturer’s installation manuals to a structurally suitable surface. The units may be mounted in any orientation.

The ERV shall have pressure taps on the unit door and onboard adjustable airflow controls for easy airflow balancing of unit.

The onboard airflow setting controls shall be factory installed and tested.

The ERV onboard control center shall have the ability to set the high and low airflow for the supply and exhaust fans independently of each airstream.

The onboard control shall have the capability to set the high and low airflow setting for the supply and exhaust fan using easy to use adjustable airflow dials that are clearly labeled outdoor air or return air and high or low for airflow setting.

The adjustable airflow setting dial shall have the capability to vary the desired airflow in infinite increments for the supply and exhaust airflows.

ERV shall have the capability to provide 220 CFM net airflow for EV Premium L on the supply air at +\- 0.8” W.G. external static pressure.

The power consumption shall be at least 1.8 CFM/watt at HVI tested and rated performance conditions. At 51 CFM, the SRE shall be at least 78%.

The ERV shall be capable of transferring both sensible and latent energy between airstreams. Latent energy transfer shall be accomplished by direct water vapor transfer from one airstream to the other, without exposing transfer media in succeeding cycles directly to the exhaust air and then to the fresh air.

Water vapor transfer shall be through molecular transport by hydroscopic resin and shall not be accomplished by “porous plate” mechanisms. Exhaust and fresh airstreams shall travel at all times in separate passages, and airstreams shall not mix. No metal separators or metal core material shall be acceptable.

Airflow through the ERV core shall be laminar over the product's entire operating airflow range, avoiding deposition of particulates on the interior of the energy exchange plate material.

Power rating of the unit shall be 120 volts and 60 Hz.

The power supply for the ERV units shall be from a 34” line cord.

Cabinet

Materials: Formed single wall insulated metal cabinet, fabricated to permit access to internal components for maintenance.

The energy recovery component shall be of fixed-plate cross-flow construction, with no moving parts.

Enthalpy core: Energy recovery core shall be of the total enthalpy type, capable of transferring both sensible and latent energy between airstreams. Latent energy transfer shall be accomplished by direct water vapor transfer from one airstream to the other, without exposing transfer media in succeeding cycles directly to the exhaust air and then to the fresh air. No condensate drains shall be allowed. The energy recovery core shall be designed and constructed to permit cleaning and removal for servicing.

Outside casing: Shall be constructed of smooth pre-painted or powder coated white 22-gauge steel, with lapped corners and zinc-plated screw fasteners.

Case walls and doors shall be fully insulated with 1 inch, expanded polystyrene foam insulation faced with a cleanable foil face on all exposed surfaces.

Access door shall provide easy access to blowers, ERV cores, and filters. Access door shall be hinged with airtight closed cell foam gaskets. Doors shall have an airtight compression seal using closed cell foam gaskets.

Door pressure taps, with captive plugs, shall be provided for cross-core pressure measurement allowing for accurate airflow measurement. Unit shall have (4) Pressure ports allow for easy airflow balancing and verification.

Unit shall have factory-supplied 6" or 8" duct collars for easy installation of ductwork to the unit.

Passive Frost Control: The ERV core shall perform without condensing or frosting under normal operating conditions (defined as outside temperatures above -10ºF and inside relative humidity below 40%). Occasional more extreme conditions shall not affect the usual function, performance, or durability of the core. No condensate drains will be allowed.

Fans

The impeller type shall be backward-curved.

Blower assemblies: Shall be statically and dynamically balanced and designed for continuous operation at maximum rated fan speed and horsepower.

Fan Motors

The supply and exhaust fans shall be electronically commutated (EC) Motors with multispeed capability as standard offering.

Unit Controls

The unit shall be capable of operating continuously or intermittently at the low airflow (adj.) setting with the ability to be switched temporarily to the high airflow (adj.) "boost" mode.

The unit shall have an internal 24VAC transformer and relay.

The ERV operates at normal airflow mode until one of the following energizes the ERV to operate on a high flow boost mode (as airflows are set during start-up and conditioning).

FILTER SECTION

The ERV cores shall be protected by a MERV-8 rated, spun polyester, disposable filter in both airstreams.

ERV shall have the capability to incorporate an optional 1" thick MERV 13 disposable pleated filters located in the outdoor air airstream.

All filters shall be accessible from the exterior of the unit.

The ERV shall have the capability to incorporate an optional 1" thick MERV 13 disposable filter (shipped loose) located in the outdoor airstream, to be installed post construction.

Requirements and standards

Complies with the UL 1812 requirements regulating the construction and installation of Heat Recovery Ventilators. Complies with the CSA C22.2 no. 113 Standard applicable to ventilators Complies with the CSA F326 requirements regulating the installation of Heat Recovery Ventilators Technical data was obtained from published results of test relating to CSA C439 Standards HVI certified.

Warrantee

The ERV core shall be warranted to be free of manufacturing defects and to retain its functional characteristics, under circumstances of normal use, for a period of ten (10) years from the date of purchase. The balance-of-unit shall be warranted to be free of manufacturing defects and to retain its functional characteristics, under circumstances of normal use, for a period of five (5) years from the date of purchase.

ERV Distribution Ductwork

Steel ductwork

Trunk ductwork shall be rigid steel, minimum 30 gage, or approved equal.

Longitudinal and transverse joints, seams and connections shall be sealed air-tight

ERV supply, return and exhaust ductwork need not be insulated. ERV intake ductwork shall be insulated with a minimum of 2” R6 wrap.

Any ducts that penetrate the residence air-barrier shall be sealed at the air barrier.

Flexible ductwork

Flexible ductwork shall be ATCO, or equal.

Flexible ductwork shall be limited to runouts of less than or equal to 10 feet. Runout duct may be flexible when 10 feet or less if installed to minimize static pressure drop.

Exterior Terminations

Exterior terminations of outside air intake and exhaust ductwork shall be to waterproof louvers, constructed with extruded stormproof blades on maximum 2” spacing to prevent water infiltration while maximizing free area. Louvers shall be heavy-duty aluminum extruded frame with mechanically locked mitered corners for exterior use. Coordinate with the Owners regarding finishes

Air Terminal Units

Supply and return air terminal units shall be Hart and Cooley, or equal.

Coordinate with the General Contractor on for color and appearance. See H-1 for the schedule.

PROGRESS

General Contractor
RON DUNN BUILDERS
Farmington, ME
https://dunnbuilders.net
207-314-3911

HVAC Design
OCTOBER ENGINEERING LLC
Palermo, ME; Sudbury, MA
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RESIDENCE
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HVAC
SPECIFICATIONS

02-14-24

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